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**The Benefits of Junk
Food People Often
Overlook**
Steve Li

**Venus as an Active
Planet**
Emily Ma

**The Role of
BioLuminescence**
Richard Wang

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The Benefits of Junk Food People Often Overlook

By Steve Li

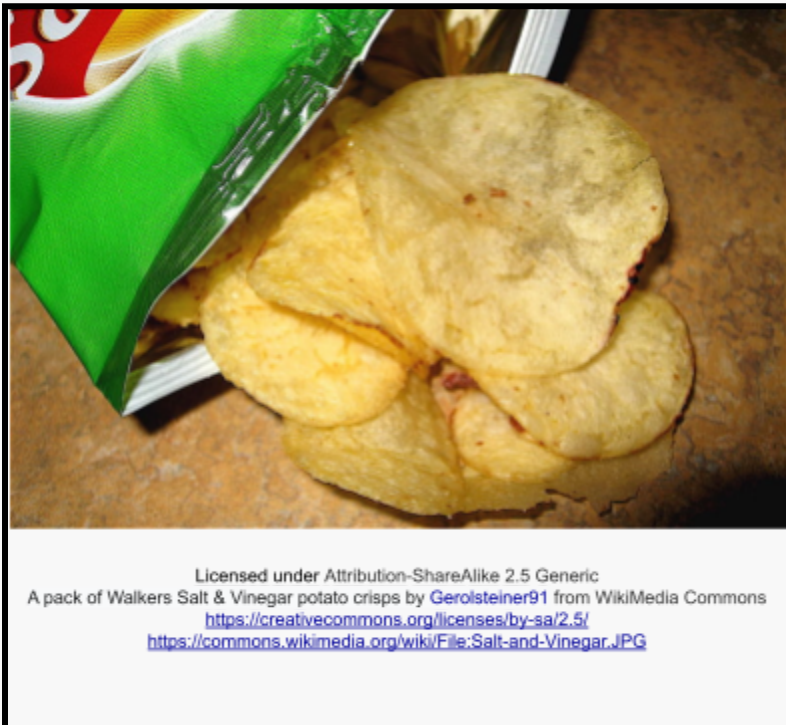
Recently, there has been a growing debate on whether junk food should be banned. While the reasons behind banning junk food are valid, many overlook the problems it will cause for several people. Along with its affordability and convenience, banning it will cause people to desire the foods more and restrict our freedom of choice.

However, some people may argue potential health problems are associated with junk food. For instance, "[We Have Kids](#)" claims, "...consuming junk food results in weight gain at a rapid pace and increases the risk of chronic illnesses..." While it's undeniable that consuming junk food can lead to weight gain and increase the risk of chronic illnesses, it's popularity is still maintained because of its convenience, taste, and affordability. This also leads us to the reason why numerous people are so fond of fast food chains like McDonald's, where they can eat inexpensive yet delicious options. Although junk food affects your health, it also has many important factors and benefits to people which absolutely outweigh the downsides of junk food.

To begin, the affordability and convenience of junk food are huge factors in why many eat it so much. A great deal of people—particularly those from low-income backgrounds—eat junk food because it is most easily accessible and they simply just cannot afford to buy the ingredients or spend precious time that they could spend working instead of going grocery shopping. The "[We Have Kids](#)" article says, "Junk food is an easy, tasty, less expensive, and convenient option for children who have busy parents." This article illustrates how junk food serves as an inexpensive and accessible alternative for children. With snacks like chips and candy typically priced at around \$2.00 or less, and being abundantly placed everywhere, junk food presents as an accessible alternative. Moreover, fast food restaurants, although not the best quality, can offer hearty

meals at a low price, catering to children whose parents might be absent during the day as well. With this evidence provided, one can conclude that junk food can benefit people in numerous ways. Junk food's convenience enables individuals to satisfy their hunger quickly and easily, regardless of their location.

Moving on, banning junk food might create the opposite effect on people. As



stated in an article in "[Should We Ban Junk Food From Cafeterias? – BBHS FOCUS](#)",

"...junk foods are craved more by kids of various schools when they are not permitted."

Many people do illegal stuff like drugs because of the fact it is banned. If we ban junk food, more harm will be caused than good. People will want to crave these foods more and obesity rates will increase even further. As said

in the article, students won't be discouraged by junk food being banned. Instead, they might find creative and different ways to eat these foods. Therefore, it is evident that banning junk food may worsen rather than improve the situation.

Finally, banning junk food can also cause trouble for students in schools. Oftentimes, students aren't satisfied with school lunch. Research shows that schools that implement a 'no junk food' policy have a higher number of kids that do not eat, either because they do not like the healthy meals served in the cafeteria or because they can't afford those options. Additionally, it is pretty common for students to complain about school lunches. An article on "[athensoracle.com](#)" written by Mohammed Salahat states

that, "...student lunches come in poor quality and affect the students negatively." Banning junk food in schools could lead to students feeling hungry and being forced to eat meals they dislike. This could result in students developing a negative perception of school and possibly even resenting it. Additionally, some students may still feel hungry even after consuming school lunches, making junk foods like chips a quick and satisfying option. Overall, banning junk food may create a negative image of schools and take away from students' enjoyment of their time there.

In conclusion, there are better solutions than banning junk food. The affordability and convenience, the potential for increased desire when not having junk food, and the importance of having freedom in what we decide to eat all argue against implementing such a drastic measure. Instead, efforts should focus on educating people to make informed decisions about their diet, rather than take away what most of them enjoy.



Advances in Computer Architecture for Machine Learning Workloads

By Eddie Zhang

The rapid growth of machine learning (ML) has driven significant advancements in computer architecture, reshaping how hardware is designed to handle intensive computational workloads. Traditional architectures, such as general-purpose CPUs, struggle to meet the demands of modern ML applications, which require high throughput, low latency, and efficient parallel processing. As a result, specialized hardware solutions have emerged, including graphics processing units (GPUs), tensor processing units (TPUs), and neuromorphic chips, each tailored to accelerate ML computations.

GPUs have become the cornerstone of ML workloads due to their ability to execute thousands of operations in parallel. Unlike CPUs, which are optimized for sequential processing, GPUs feature thousands of cores designed for simultaneous computations, making them ideal for training deep neural networks. The evolution of GPU architectures, such as NVIDIA's Ampere and Hopper series, has introduced features like mixed-precision computing and tensor cores, which further enhance ML performance while improving energy efficiency. These innovations allow researchers and developers to train models faster and at lower computational costs.

Beyond GPUs, TPUs have gained prominence as dedicated accelerators for ML workloads. Developed by Google, TPUs are designed specifically for deep learning tasks, offering optimized matrix multiplication and lower power consumption compared to traditional processors. TPUs excel in handling large-scale ML models, particularly in cloud-based environments where speed and efficiency are critical. By integrating TPUs into data centers, companies can significantly reduce the time required for training and inference, enabling real-time AI applications.

In addition to GPUs and TPUs, field-programmable gate arrays (FPGAs) and application-specific integrated circuits (ASICs) are being increasingly used for ML acceleration. FPGAs provide flexibility, allowing developers to customize hardware configurations for specific ML tasks. Their reconfigurable nature makes them ideal for real-time applications, such as autonomous vehicles and edge computing. Meanwhile, ASICs are designed for ultra-efficient ML processing, offering unparalleled speed and power efficiency for tasks like speech recognition, natural language processing, and computer vision.

Neuromorphic computing represents a cutting-edge advancement in computer architecture, inspired by the structure and function of the human brain. These chips, such as Intel's Loihi and IBM's TrueNorth, use spiking neural networks to process information in a highly efficient and biologically plausible manner. Neuromorphic processors have the potential to revolutionize edge AI by enabling ultra-low-power inference, making them suitable for embedded systems and Internet of Things (IoT) devices.

As ML workloads continue to evolve, future advancements in computer architecture will focus on improving energy efficiency, reducing latency, and scaling performance. The integration of AI-driven hardware design, chiplet-based architectures, and quantum computing could further push the boundaries of ML acceleration. By continuously innovating, engineers and researchers can ensure that hardware keeps pace with the growing demands of artificial intelligence, unlocking new possibilities for industries ranging from healthcare to autonomous systems.

Crystal Engineering: Designing Functional Materials

By Brandon Wang

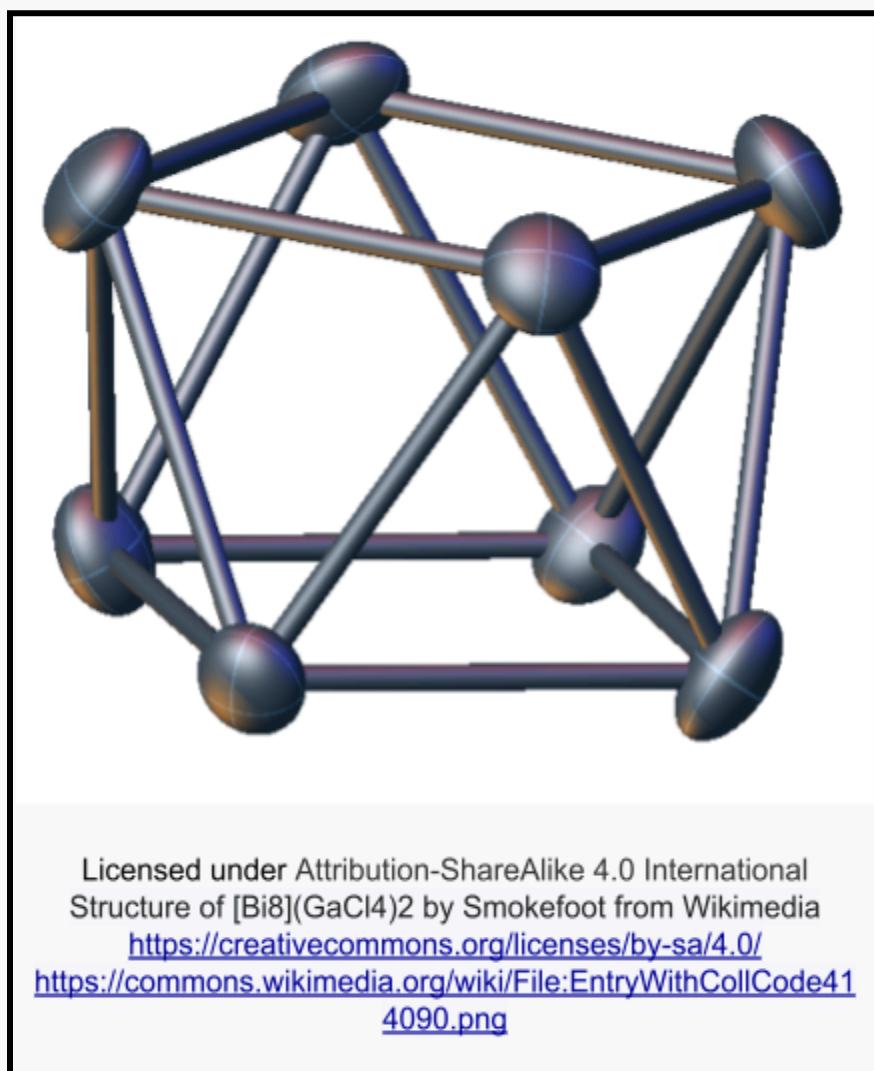
Crystal engineering is the practice of developing and creating crystalline solids that exhibit desired properties by controlling molecular structures. It is a combination of chemistry, physics, and materials science aimed at creating functional materials that have uses in pharmacy, energy storage, and electronics. The topic relies on the knowledge of non-covalent interactions between molecules, like hydrogen bonding and van der Waals interactions, which control the manner in which molecules organize themselves in a crystal lattice.

Supramolecular synthons, repeating units of intermolecular interactions used to predict and design crystal structures, are one of the key concepts. Polymorphism, where one compound can form multiple crystal structures with different properties, is another key concept. Control of polymorphism is extremely important in the pharmaceutical field, as different crystal forms can affect the solubility and effectiveness of a drug.

Crystal engineering is applied in a variety of ways. In medicine, it enhances drug stability and solubility by the use of co-crystals. In materials science, it allows for the fabrication of organic semiconductors and photonic crystals. Engineered crystals are employed in gas storage and catalysis using porous materials such as metal-organic frameworks (MOFs). They also find applications in energy storage in batteries and sensors that sense environmental changes.

In spite of its promise, crystal engineering is confronted with serious challenges. Crystal structure prediction is still challenging owing to intricate intermolecular interactions, and scaling up for industrial applications may be problematic. Moreover, some of the engineered crystals may be unstable under specific conditions. In

conclusion, crystal engineering is a successful strategy for the design of materials with tailor-made properties. Through the control of molecular interactions, scientists can create new solutions to medical, energy, and technological problems. With ongoing development, the field has even more revolutionary uses.



Why Do People Yawn?

By Mary Liang

Throughout my research, yawning has mostly revolved around the theory that it helps keep the brain awake and in addition as a way to make one more alert to their surroundings. When one yawns, the body's airways expand fully, and the surrounding muscles near the pharynx stretch, or tense. Yawning is considered to be mostly involuntary and most people believe that it is a reflex. It is also possible that yawning is contagious, especially with close acquaintances.

One of the theories is that animals yawn is the most common reason that one thinks up daily, from tiredness, boredom, waking up, and stress. This is one of the most popular reasons since most people can relate to it on a conscious level. Another theory is that yawning helps keep the brain awake and stimulate heart rates, it may also increase blood flow to the brain which increases the focus level. It may also bring in that yawning equalizes the pressure in one's ears, relating to the former statement. Animals yawn due to the need to be able to synchronize the mood of the group, and may also regulate its core temperature.

However, yawning is not always caused by those reasons. For example, when one is in an airplane as it changes elevation rapidly, one might find oneself yawning due to the air pressure caused by the altitude shift. Yawning To summarize, yawning does not have certain reasons, but the limited research points toward the theory that yawning is caused by sleepiness, boredom, and stress.

An Unconscious Habit, With its Good and Bads

By Jerry Yang

A habit we've all subconsciously had without ever questioning why. Our actions seemingly contrast, as pain and pleasure come from the scratching of an itch from the common mosquito bite or a simple infection or rash.

Scratching does lead to self-inflammation, causing the body to go into overdrive on certain organs and facial parts. The act of scratching causes the activation of mast cells, leading to signals released that kick off the reddening and swelling you see on parts of your body after scratching an itch, causing more trouble than good.

Research has been done on mice, with controlled groups made up of either all



mice with the ability to scratch itches around the ear, or those that can't feel itches or aren't able to reach them. Mice without the need or ability to scratch itches had quick relief, as inflammation never occurs and the original cause of the itch quickly disappears.

The good news is, itching can lead to a heightened immune response against bacteria on skin. Because of the inflammation, the body is able to react more often and accordingly, being

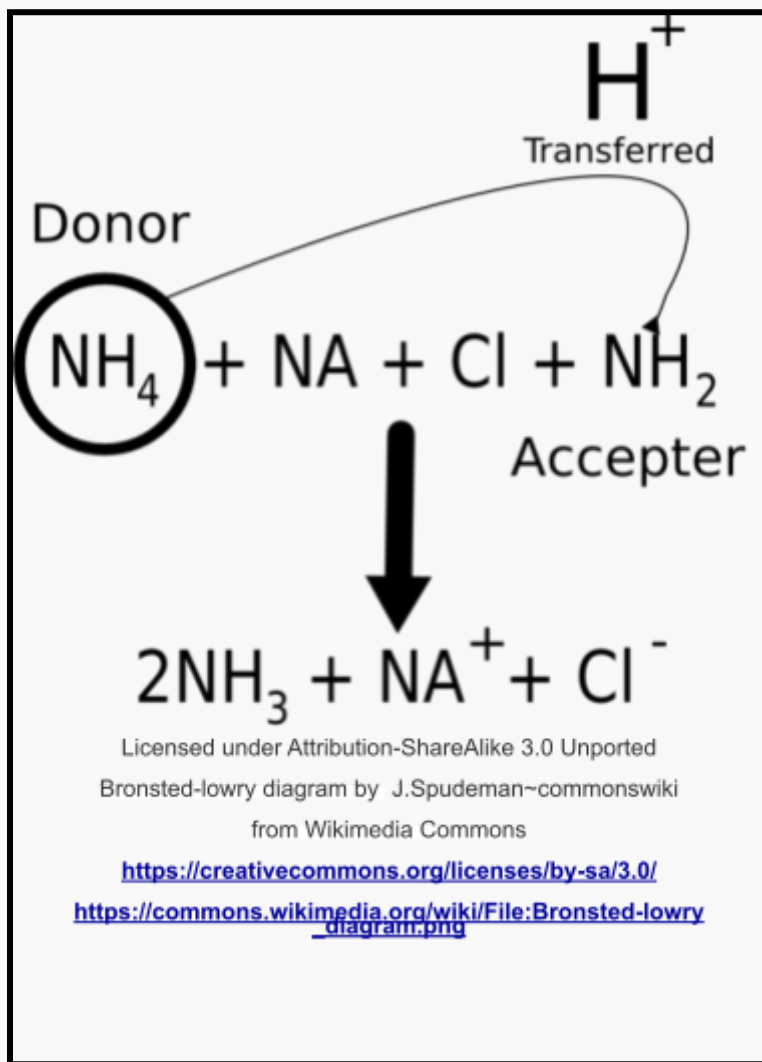
exposed to more bacteria. A possible consequence of this is breaking the protective barrier that prevents bacteria from entering the body. Piercing or tearing off skin will lead to worse infections, so it is best to be cautious.

Acids and Bases

By Audrey Wu

Acids and bases are fundamental concepts in chemistry, characterized by their distinct properties and behaviors. Acids are substances that release hydrogen ions (H^+) when dissolved in water, while bases release hydroxide ions (OH^-). These ions are responsible for the distinctive characteristics of acids and bases. Acids are often sour to taste, can conduct electricity in aqueous solutions, and turn blue litmus paper red. Bases, on the other hand, tend to be bitter and slippery to the touch and turn red litmus paper blue. These properties are not only crucial for understanding chemical reactions but also for a variety of applications in daily life, such as in cleaning products or in biological systems.

One of the most widely known theories for describing acids and bases is the Arrhenius theory. According to this theory, an acid is a substance that increases the concentration of hydrogen ions in aqueous solution, whereas a base increases the concentration of hydroxide ions. For example, hydrochloric acid (HCl) dissociates



in water to release H^+ ions, while sodium hydroxide (NaOH) dissociates to release OH^- ions. However, the Arrhenius definition has its limitations, especially when it comes to non-aqueous environments, leading to other models such as the Bronsted-Lowry theory, which defines acids as proton donors and bases as proton acceptors, regardless of the solvent used.

Another important aspect of acids and bases is their ability to neutralize each other in a chemical reaction. When an acid and a base react, they form water and a salt in a process known as neutralization. This is a crucial reaction in many industrial and environmental processes, such as in water treatment or the manufacture of fertilizers. The pH scale, which ranges from 0 to 14, is a measure of the acidity or basicity of a solution. A pH of 7 is considered neutral, with values less than 7 indicating an acidic solution and values greater than 7 indicating a basic solution. The pH scale is a valuable tool for measuring and controlling the concentrations of acids and bases in different environments.

In biological systems, acids and bases play essential roles in maintaining homeostasis. For example, the human stomach produces hydrochloric acid to aid in digestion, while the blood maintains a slightly basic pH to ensure proper enzyme function and overall metabolic balance. The body employs various buffering systems to regulate pH levels, preventing harmful fluctuations that could disrupt normal physiological processes. Understanding acids and bases is not only fundamental to chemistry but is also crucial in medicine, environmental science, and many other fields. Their influence extends far beyond the laboratory, making them a central part of both scientific study and everyday life.

Venus as an Active Planet

By Emily Ma

For a large part of the history of exploration, Venus was thought to be Earth's "sister planet" but a dead, inhospitable planet with runaway surface temperatures and a dense atmosphere filled with sulfuric acid and other toxic gases. Yet, as time passes and discoveries continue, such assessments are short-lived—there seems to be much more



life and activity on Venus than scientists previously thought. For example, discoveries about volcanism support the fact that a team recently studied the legacy of the Magellan mission to Venus, which was commissioned by NASA in 1990. After examining the findings of Magellan's radar in 1991, it was determined

that surface properties of Venus changed over time, suggestive of volcanism occurring several decades ago. Therefore, now, Venus is classified as one of three volcanic worlds within the solar system, along with Earth and Io, a moon of Jupiter. This classification not only allows scientists to reassess how much volcanically active Venus was over time, but it suggests some form of internal heating/activity is still happening on Venus, regardless of its dead shell appearance. Therefore, if volcanism is an observable phenomenon that scientists could always refer to, this could be a new avenue for assessing life detection possibilities on Venus.

Yet where such hostile surface temperatures (over 460 degrees Celsius) and atmospheric pressures (over 90 times Earth's) exist, one would think life improbable; however, scientists hypothesize microbially in the temperate cloud layers. Since 2020, scientists have discovered that phosphine gas resides in the atmosphere of that planet; however, this is a gas usually associated with biogenic processes. Although detection is still in question, it has led to similar complication studies with extremophile life forms who may exist in acid rain. For example, volcanoes exist on Venus and could be responsible for the presence of phosphine as well as other biosignatures. Thus, it appears that Venus is not as dead as thought, and further studies should be conducted. In addition, NASA has two missions, VERITAS and DAVINCI+, and the European Space Agency's EnVision mission, which will achieve atmospheric and topographical mapping better than ever before. What scientists hope to find matters to Earth and potentially life outside Earth's atmosphere. By analyzing volcanic activity and atmospheric components, scientists can determine what Earth needs to avoid with evolution and growth. Where once Venus was thought to be an uninhabitable wasteland, it is now becoming acknowledged as a habitable place with much to offer regarding the origins of life and sustainability beyond Earth.

Kirchhoff's Laws: Understanding the Current and Loop Rules

By Kenny Wu

Electricity powers nearly every aspect of modern life, from household appliances to advanced technological systems. To understand how electrical circuits function, scientists and engineers rely on fundamental principles like Kirchhoff's Laws. These laws, developed by Gustav Kirchhoff in the 19th century, explain how electric current and voltage behave in circuits. Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL) provide a framework for analyzing electrical systems, ensuring that circuits function properly and efficiently.

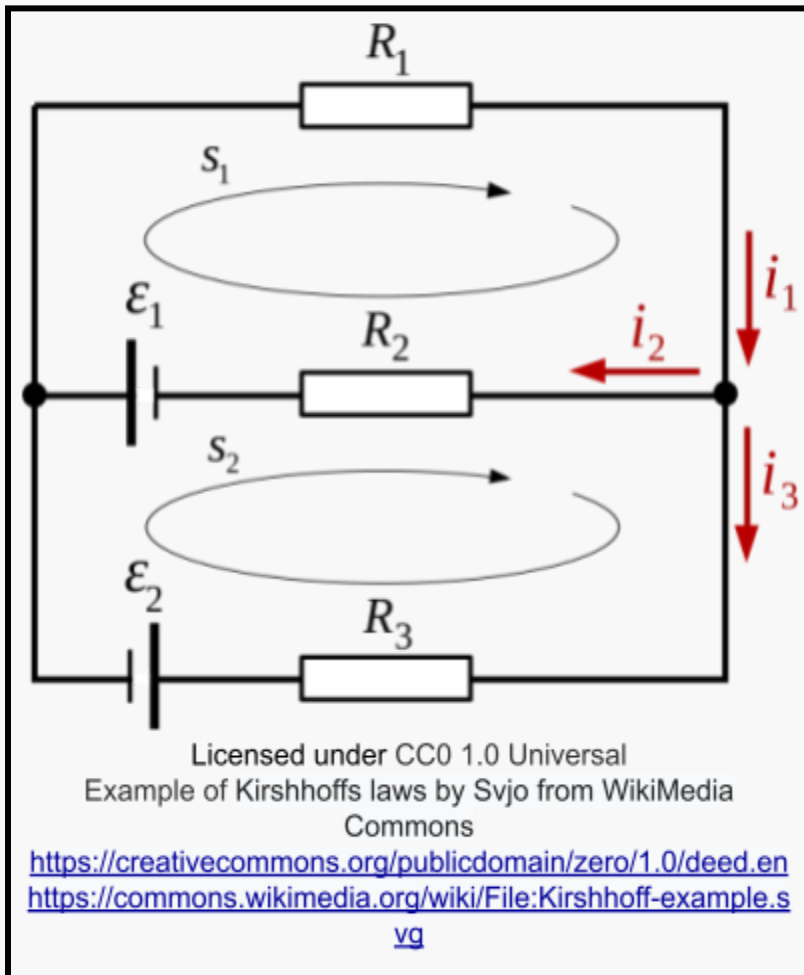
Kirchhoff's Current Law, also known as the Current Rule, states that the total current entering a junction in a circuit must be equal to the total current leaving it. This principle follows the idea that electric charge cannot be created or destroyed within a circuit. At any point where wires connect, the electric current that flows in must be the same as the current that flows out. This law helps engineers determine how electrical current is distributed in circuits, particularly in systems with multiple pathways. Without this rule, it would be difficult to predict how electricity flows through complex networks.

Kirchhoff's Voltage Law, also called the Loop Rule, explains how voltage is distributed in a circuit. It states that the total voltage supplied by power sources, such as batteries, is completely used up by the components in the circuit. As electric current flows through a closed loop, it encounters resistance in components like resistors and light bulbs, causing the voltage to decrease. However, the total energy lost by these components always equals the energy supplied, ensuring that no energy is lost or gained unexpectedly. This law is essential for designing circuits that operate efficiently and safely.

Both Kirchhoff's Current Law and Voltage Law play a crucial role in electrical engineering and physics. They are used to analyze circuits, troubleshoot electrical problems, and design systems ranging from simple household wiring to complex electronic devices. Without these principles, engineers would struggle to predict and control how electricity flows in different situations. These laws also serve as the foundation for more advanced circuit analysis techniques, allowing scientists to build

more sophisticated electrical systems.

In conclusion, Kirchhoff's Laws are essential for understanding how electrical circuits work. The Current Rule ensures that electric charge is conserved at junctions, while the Loop Rule guarantees that energy is balanced in a circuit. Together, these principles provide the foundation for designing and analyzing electrical systems of all sizes. Whether in household electronics or large-scale power grids, Kirchhoff's Laws help ensure that electricity



flows smoothly and safely, making them a cornerstone of electrical science.

The Role of Bioluminescence in Deep Sea Ecosystems

By Richard Wang

Bioluminescence, the production and emission of light by living organisms, is a fascinating phenomenon that plays a crucial role in the survival of many species in the deep sea. The deep ocean is a largely unexplored and extreme environment,



characterized by complete darkness, high pressures, and cold temperatures. Despite these challenges, many marine organisms, from bacteria to fish, produce their own light through bioluminescence. This essay explores the role of bioluminescence in deep-sea ecosystems and how it aids in communication, predation, and survival.

In the deep sea, where sunlight does not penetrate, bioluminescence serves as an important tool for communication. Many species, such as deep-sea fish and squid, use bioluminescence to communicate with each other. For example, certain species of squid produce light to attract mates, signaling their readiness to reproduce. Similarly, firefly squids use bioluminescence to create patterns of light for social interactions, helping individuals of the same species identify each other in the pitch-black environment. This form of communication is essential for survival, allowing individuals to locate one another in a vast and dark ocean.

Bioluminescence also plays a key role in predation and hunting. Some deep-sea predators use their own bioluminescence to attract prey. For example, the anglerfish has a bioluminescent lure on its head, which it uses to attract smaller fish toward its mouth. Similarly, species such as lanternfish and cookiecutter sharks use their bioluminescence to camouflage themselves or deceive other animals, aiding in successful hunting. These bioluminescent adaptations enhance the predators' ability to find food in the darkness of the deep ocean, where traditional vision is ineffective.

On the other hand, bioluminescence also serves as a form of defense for many species. Some animals, like squid and jellyfish, release clouds of bioluminescent material to confuse or distract predators. This tactic, known as counterillumination, allows the prey to escape by camouflaging itself against the faint light from above. Additionally, certain fish use bioluminescence to create a dazzling display that disorients predators, allowing them to flee before being caught. This defensive mechanism is a critical adaptation in a predator-rich environment where survival depends on avoiding danger.

Furthermore, bioluminescence has been shown to have an important ecological role in the deep-sea food web. Organisms that produce light, such as plankton, serve as a food source for other creatures. The energy generated by bioluminescent organisms can be passed along the food chain, supporting complex ecosystems. Some species of bioluminescent bacteria form symbiotic relationships with larger animals, providing them with light in exchange for nutrients. These mutually beneficial interactions help maintain the delicate balance of deep-sea ecosystems.

In conclusion, bioluminescence is a crucial adaptation for survival in the deep-sea environment. It aids in communication, predation, camouflage, and defense, allowing species to thrive in one of the most extreme habitats on Earth. As scientific exploration of the deep ocean continues, bioluminescence remains an area of great interest, offering insight into the remarkable adaptability of life in the deep sea.

Tragedy of the Commons

By Owen Chen

The Tragedy of the Commons is a concept in environmental science and economics that describes how individuals, acting in their own self-interest, can lead to the depletion of shared resources. This idea was first introduced by ecologist Garrett Hardin in 1968, emphasizing the dangers of unregulated resource use. The phenomenon



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occurs when multiple individuals exploit a common resource without restriction, ultimately resulting in its degradation or collapse. It applies to various ecological and economic systems, making it a critical issue in sustainability and resource management.

A common-pool resource is a type of resource that is available to multiple individuals but is limited in supply. Examples include fisheries, forests,

freshwater sources, and the atmosphere. These resources are characterized by their rivalrous nature, meaning that consumption by one individual reduces the amount available to others. Since no single person owns these resources, there is little incentive for individuals to conserve them, leading to overuse and eventual depletion. This process illustrates the Tragedy of the Commons, where short-term individual gain leads to long-term collective loss.

One of the most well-documented examples of this phenomenon is overfishing. In marine ecosystems, fish populations can only regenerate at a certain rate. However, if too many fish are harvested without regulation, populations decline to unsustainable levels, disrupting the food chain and affecting biodiversity. Similarly, deforestation occurs when trees are cut down at a rate faster than they can be replenished, leading to habitat destruction, reduced carbon sequestration, and loss of biodiversity. Another modern example is climate change, where excessive carbon emissions from industrial activities contribute to global warming, despite the collective consequences for all nations.

Scientific approaches to mitigating the Tragedy of the Commons involve policy interventions and economic mechanisms. Governments and international organizations implement regulations, such as fishing quotas, emissions limits, and protected areas, to prevent resource exhaustion. Economic strategies like carbon pricing, cap-and-trade systems, and subsidies for sustainable practices incentivize responsible resource use. Additionally, community-based resource management, where local populations take collective responsibility for conservation, has proven effective in some regions.

In conclusion, the Tragedy of the Commons is a critical issue in environmental science, demonstrating how unregulated resource use leads to long-term depletion. Scientific research and policy interventions are essential for addressing this problem and ensuring the sustainability of shared resources. By integrating economic, ecological, and regulatory solutions, societies can mitigate the effects of the Tragedy of the Commons and promote a balance between human needs and environmental conservation.

How 3D Printing Can Revolutionize the World

By Riley Lee

Have you ever wondered how 3D printing can change the world? This innovation also known as additive manufacturing have the ability to change industries by opening up many benefits including faster production, flexibility, and waste reduction. 3D printing is a method of creating objects layer by layer using a computer created design. By creating objects or structures layer by layer from digital models, it allows new and creative ideas in fields from aerospace to construction. 3D printing also allows many groups and individuals to express, create, or show their ideas or solutions to everyday or long term problems such as storing objects.

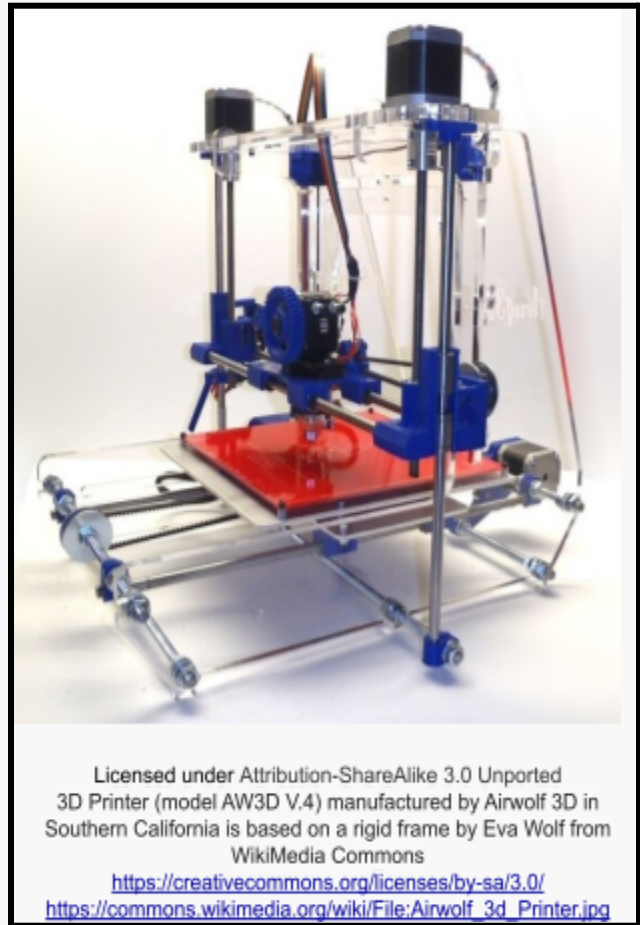
In the aerospace industry, 3D printing is used to create stronger and lighter objects like engines and turbines. An example of 3D printing in the aerospace industry is GKN Aerospace, as they have used this technology to produce a 3D printed compressor case for Rolls Royce UltraFan engine. The goal of the UltraFan engine is to lower emissions through better fuel efficiency. This example helps show how 3D can help in making aerospace technology more green. In addition, 3D printing allows a fast phase of prototyping as it gives engineers and designers the ability to test and refine the new designs quickly, which is super important in an industry like aerospace.

In the medical field, 3D printing contributes by creating medical tools, custom prosthetics, and bioprinting for both the patients and workers. Bioprinting is using layered biomaterials combined with a patient's own cells to create bioartificial organs. As 3D printing evolves, so will bioprinting and this will influence the savings of a thousand lives by providing transplant patients with personalized, lab grown organs. As 3D printing produces everyday medical equipment such as medical clamps and dental tools. The

dental industry benefits from the functions of 3D printing because of its ability to produce precise and important components. With 3D printing in the medical field, it improves patients' results and the efficiency of the field's work.

In the construction industry, 3D printing is used as a method to speed up the building time and decrease the cost of the work. With large scale 3D printers, the foundation or the basic structure of houses can be built quicker and more efficiently. An example of 3D printing in the construction industry is in Georgetown, Texas, Danish architecture firm BIG and American construction tech company ICON are working on a project called the Wolf Ranch project to create the largest community of 3D printed homes.

In conclusion, 3D printing's ability to create new and unique designs while allowing increased production and reduced waste is laying the beginning of a new start for innovation. As technology develops, its influence will too and expand across a variety of industries. Whether the industry is automobiles, medical, or construction, 3D printing has the power to revolutionize the world.



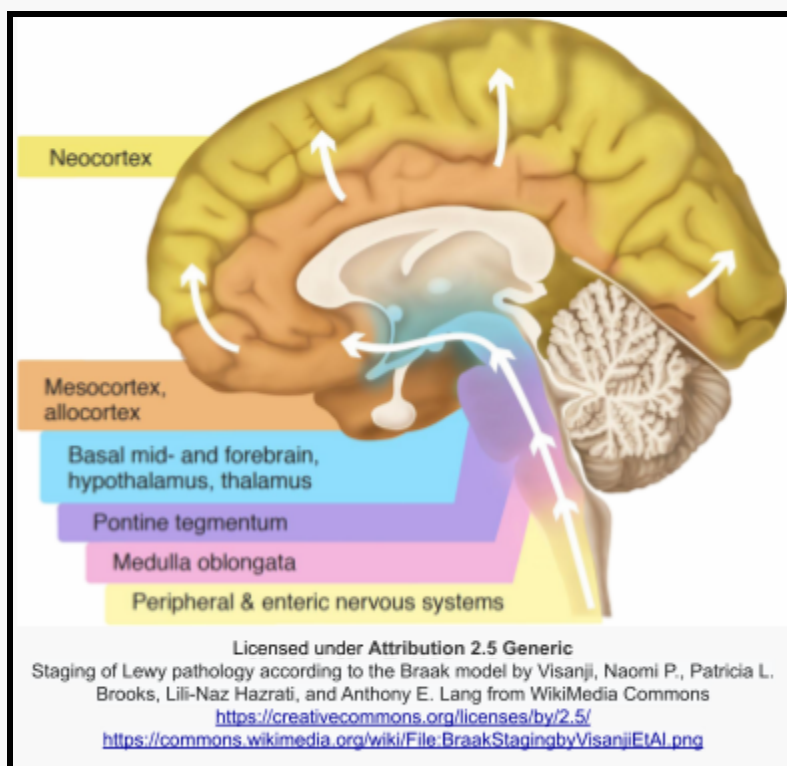
Understanding Neurodivergence

By Denise Lee

Neurodivergence represents normal brain operating patterns that deviate from what is defined as typical or neurotypical. This term encompasses conditions such as autism spectrum disorder (ASD), attention deficit hyperactivity disorder (ADHD), dyslexia, and obsessive-compulsive disorder (OCD), among others. The neurodiversity movement supports full neurodivergent individual acceptance by acknowledging their both positive traits and difficulties in addition to rejecting any objective to fix these differences.

Neurodivergence develops as a result of physical brain variations, together with

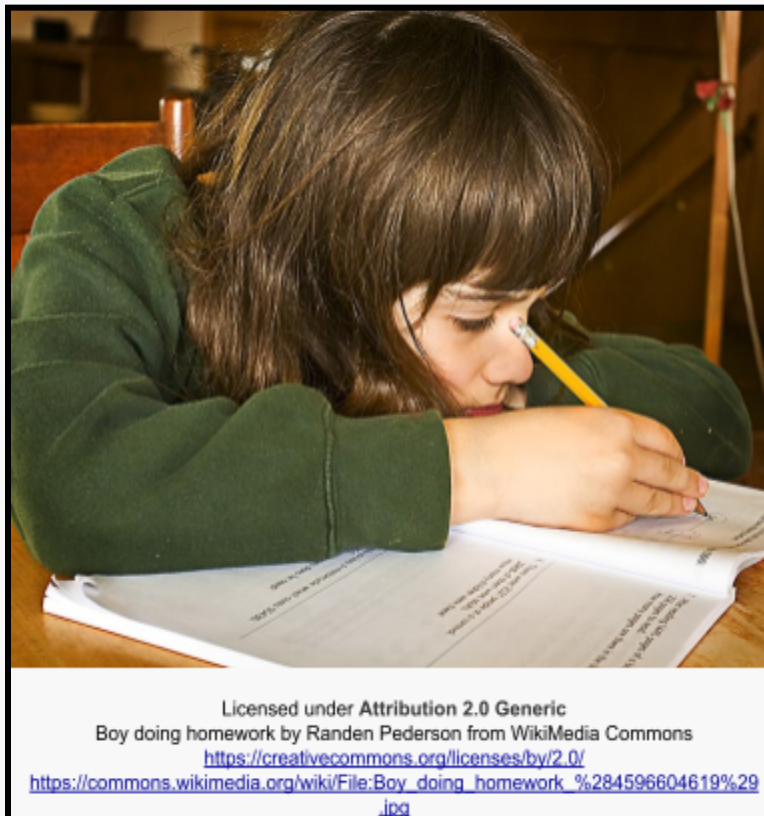
neural pathways variations and alterations in chemical signals. Studies demonstrate that ADHD patients experience reduced levels of dopamine, which makes it difficult for them to stay focused and control instincts. Neuroimaging research shows autistic individuals display stronger brain connections in pattern recognition parts of the brain that support logical thinking, although these networks might show different



functions for social processing. The brain differences between these conditions shape how neurodivergent individuals think, process information, and behave, making each condition distinct from the others.

Neurodivergence creates unique impacts on how people think and learn, together with their social capacities. Those who have ADHD face executive function challenges yet demonstrate superior capabilities in creativity together with problem-solving abilities. Autistic people experience difficulties with social communication yet possess remarkable abilities to examine details extensively within their areas of expertise. Recognizing neurodivergence between people enables modern societies to eliminate outdated stereotypical beliefs, thus embracing diverse perspectives.

All in all, acceptance of neurodiverse features in individuals leads to a society that embraces inclusivity together with understanding. Our society should work towards accommodating intellectual diversity instead of requiring people to match typical neurological patterns because cognitive diversification deserves acknowledgment and supportive environments need to be created. We need to approach a future where diversity will be understood beyond physical appearance to include thinking and perceiving methods.



Human Heredity and Pedigrees

By Brandon Pian

Human heredity is the study of how genetic traits are passed from one generation to the next. Our genetic information is stored in DNA, which is organized into chromosomes. Each person inherits 23 pairs of chromosomes, one set from each parent. These chromosomes contain thousands of genes, which determine traits such as eye color, blood type, and even predisposition to certain diseases.

Mendelian inheritance, discovered by Gregor Mendel, explains how traits are inherited through dominant and recessive alleles. A dominant allele only needs one copy to be expressed, while a recessive allele must be inherited from both parents to be visible in an individual. However, not all traits follow simple Mendelian patterns. Some traits are influenced by multiple genes which is known as polygenic inheritance, while others may involve environmental factors.

To study inheritance, geneticists use pedigrees, which are family tree diagrams that track the transmission of traits over multiple generations. In a pedigree, males are represented by squares, females by circles, and shaded symbols indicate individuals who exhibit the studied trait. By analyzing patterns in a pedigree, scientists can determine whether a trait is inherited in a dominant, recessive, or sex-linked manner.

Pedigree analysis plays a crucial role in medicine, especially in predicting the likelihood of genetic disorders such as cystic fibrosis, Huntington's disease, and hemophilia. This knowledge allows for genetic counseling, helping families make informed health decisions.

Understanding heredity and pedigrees not only provides insight into genetics but also helps us appreciate the complexity of human inheritance and evolution.

What is a MERN Stack?

By Aidan Hong

Many websites offer a wide range of services, from allowing you to watch movies, buy food, and more. However, the technology behind such services is more complex than many expect. Some developers will utilize MongoDB, Express.JS, React.JS, and Node.JS(MERN) to create such an advanced website, taking care of both the front end and back end to create a full-stack application.

The first part of a MERN stack is MongoDB, which is used for back-end functions. MongoDB is a NoSQL database or a database that doesn't have any relations. A relation in a database is when data is linked to other data in another table. This means that data stored in a MongoDB database will not be linked to other data in that same database in any shape or form. Typically, most will use MongoDB to store usernames and password hashes for authentication. Password hashes are stored instead of passwords because password hashes are hard to break and typically need brute force tools like hashcat to crack. Most developers will typically use MongoDB, but if the developer needs a relational database and SQL, a popular alternative is PostgreSQL.

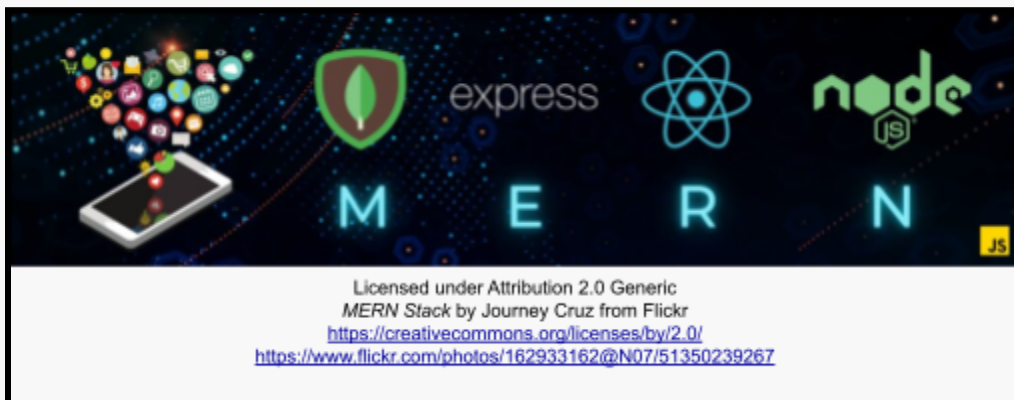
The second part of a MERN stack is Express.JS. Express is used to handle HTTP requests and responses, as well as manage data and implement logic for the server, where the back-end program is usually hosted. Express is used when the user requests data: express will handle which route will handle the data and respond with the appropriate data. Express is popular: chances are, if you click "inspect element" and look in the network tab, most data that you request will use express!

The third part of a MERN stack is React.JS. React is typically used for the front end. React offers a framework developers can use to create the front end of a webpage. The front

end is typically used to show the content to the user, as well as show data from the backend in a presentable format, after all, it isn't fun to read raw JSON. React uses JSX components, which include aspects of both JS and HTML. Their syntax is almost identical to HTML! While React.JS is great, there are also many other alternatives to React.JS, such as Next.JS, Vite, Angular, and more.

Finally, the last part of a MERN stack is Node.JS. Node.JS is used to create the overall backend. This is often used to run the actual code: while Express handles the logic and routes data appropriately, Node.JS will run the data and return the actual data. Additionally, Node.JS is also used to call API's. API's are requests to other servers or websites, and can be used to expand the website's functionality without having to write much code: the only code needed is to call the API and format the data sent and received! There are also many other alternatives to Node.JS, such as Flask, Java, and Ruby on Rails.

All of these technologies combine to create a MERN Stack. MongoDB is used for authentication as well as to store data, while Express is used to handle logic and route data accordingly. React is used to create a great frontend while Node.JS is used to handle backend programming and works with express to return data to the user. Combined, we get the powerful and stunning websites we all love and enjoy using.



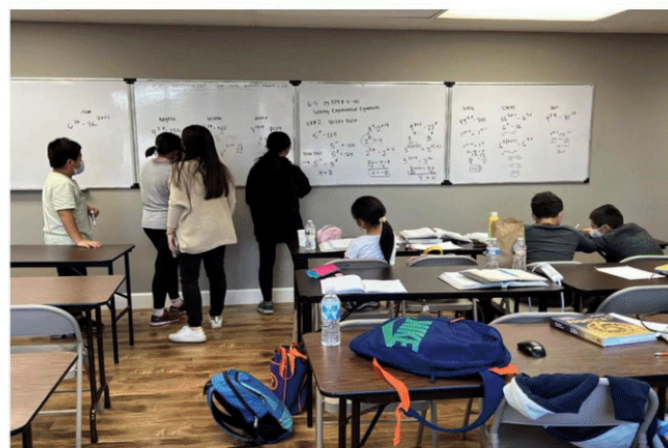
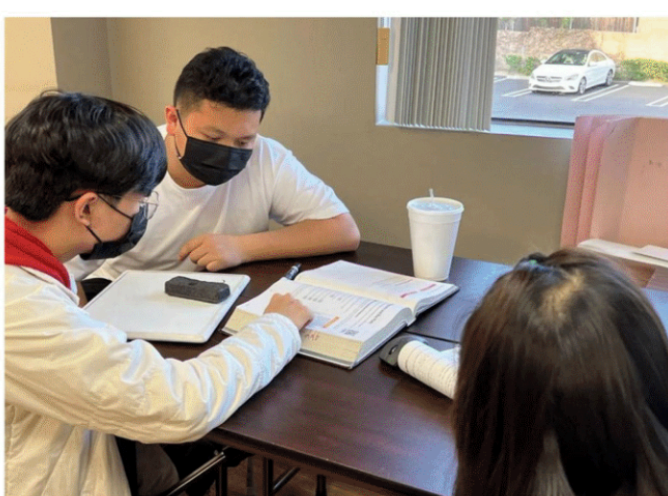
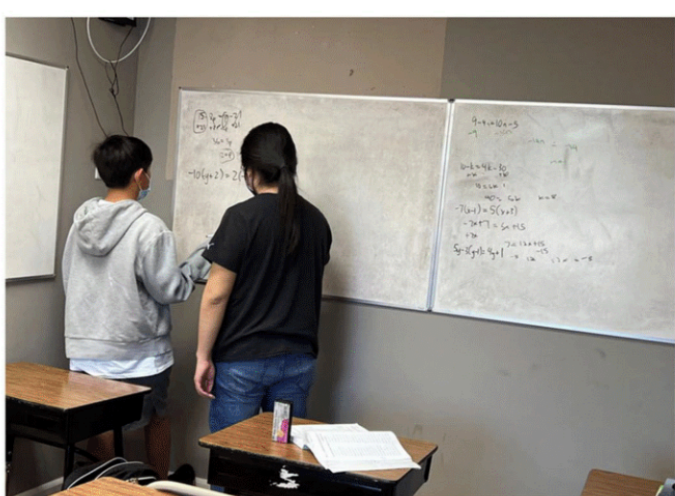
MLB ABS Challenge System

By Arick Hong

If you have ever played baseball, you'll figure out quickly how troublesome umpires are. Umpires often fumble critical strike or ball calls, sometimes costing a team the game. Umpires such as Angel Hernandez are famously known for blowing strike and ball calls that are not even close to the strike zone. However, this year in spring training, the automatic ball strike call system or ABS is being tried out to solve this issue. This article will explore how it works, and whether it is good or not.

The strike zone in baseball is determined by the player's height. All the players are first measured without shoes to create an individualized strike zone. Using high tech cameras that track the baseball, ABS then uses some code to determine if the ball is touching the strike zone or not. When a player or coach challenges the pitch, the ABS has the result for the call already. ABS is very accurate too, being able to determine calls less than 0.5 inches away from the strike zone. The umpires then overturn the call or keep the call depending on what the ABS says. These calls can really help change the pace of the game by helping the batter or the pitcher.

While ABS can be helpful, umpires are worried that it may take their jobs, causing some controversy. Ultimately, ABS will help revolutionize the game of baseball with its high efficiency technology, eventually replacing regular umpires and making ball strike calls perfect from now on.



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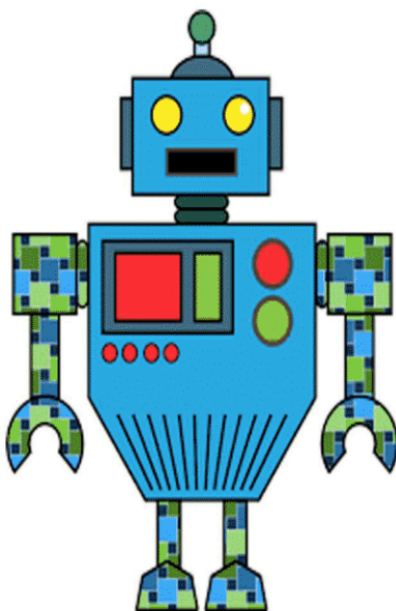
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